Paper Review: Strings from Almost Nothing —C. Cheung et. al

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1. **Bibliographical information:**
   C. Cheung, G. Remmen, F. Sciotti, M. Tarquini, Strings from Almost Nothing, Phys. Rev. Lett. 136, 251601 (2026), [arXiv:2508.09246].

2. **Main Claim:**
   Under tree-level amplitudes, people don't have to introduce extra dimensions, with world-sheet, with SUSY. The string theory's result (Veneziano amplitude, Virasoro-Shapiro amplitude) can emerge just from some fundamental assumptions, causality, unitarity, ultrasoft high-energy behavior, and minimal zeros. So string theory is not an artificially arbitrary product, but an inevitable result of nature.

3. **Why is this interesting?:**
   String theory turned from flourish to dying, since it seems not match to nature, and some results are not checkable. But it is such a beautiful model that some people don't want to totally give it up. So an alternative is to logically prove the result is true. This method, called bootstrap, is giving string theory and others in the similar situations a new life.

4. **Guess how the result was obtained:**
   The authors might start from some basic assumptions that almost every physicists accepted, locality, unitarity, Lorentz invariance, along with some special assumptions, like the ultrasoft high-energy behavior. Since the ultrasoft behavior states that the amplitude vanishes rapidly at infinity, which is a common condition in physics, and naturally the boundary term of the contour integral can vanish. By continuing to the complex plane, inversely from the value of the integral, they can constrain the singular points to some specific areas on the complex plane so that their residues can contribute to a correct integral value. From introduction, those residues (as polynomials of t) must vanish at specific ts (Regge zeros). Then use bootstrap to construct a minimal space with those singular points, and collapse to Veneziano and Virasoro-Shapiro amplitudes.

5. **My guess vs. actual paper:**
   The literature uses the uniqueness of analytical continuation to show the residues vanish. It also uses calculation as I guessed. They calculate the value of the integral without using residue, and compare with the result in terms of residues to show that residues must vanish. However, it doesn't show the position of singular points on the complex plane. Instead, it gives the position of Regge zeros on the real t axis. After this, the literature

shows the derivation of the uniqueness of string theory. It uses the logarithm method. (This is interesting because my unfounded association to the authors' adjacent work, the inverse problem of EFT, in which they also use logarithm method.) It derived the mass square is proportional to spin, and another linearity of Regge trajectory, as an important result of the paper. Then the residues degenerates to the result (level truncation) of a previous paper, which solution correspond to Veneziano amplitude. For non-planar four point scattering, the result leads to another previous paper, which solution correspond to Virasoro-Shapiro amplitude.

6. **Regime of Validity:**
   The results based on tree-level approximations, which means strong coupling cases may beyond the regime of validity. The authors also admitted that in their minimal zeros assumption, the exclusion of other zeros is not conservative. But before they used the minimal zeros, the constraints of amplitude still works, it is just cannot determine the amplitude. While if the amplitude is not determined to the string theory amplitude, this bootstrap result is not valid.

7. **What would you do next?:**
   The most natural next step is to generalize to more points scatterings or other non-planar geometries, as stated in the paper. The final vision is to find a general description for all cases if possible.
   Next, I want to ask, what in the string theory correspond to the assumptions in this paper to form the string amplitude receptively?
   If there are extra zeros, what will happen? What amplitude we can get?
   Next, understand in loop-level can people still bootstrap the string amplitudes? If not, why, does that means people still have to use string theory to get those amplitudes or string theory is not the truth on that level?

8. **Mind Palace Entry:**
   This paper starts from causality, locality, assume amplitudes vanish at infinity and no extra zeros beyond necessity, using complex analysis, proved the residues collapse to the forms that correspond to string amplitudes, and generalized to the four-point closed-string and five-point open-string amplitudes.